Distillation Unit



Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Distillation Plant

What it is used for

A distillation unit separates components of a liquid mixture based on differences in volatility by heating, vaporising, and condensing the vapour. Proper setup, controlled heating, cooling-water flow, and safe collection help improve separation while reducing exposure and equipment hazards.

Main parts

- Distillation flask or feed vessel
- Heating mantle, hot plate, or approved heater
- Distillation head and thermometer
- Condenser and cooling-water connections
- Receiving flask or collection vessel
- Clamps, stands, tubing, and protective equipment

Before you start

1. Review the experiment or process procedure, safety data sheets, and the hazards of the liquid being distilled.
2. Confirm that the glassware is compatible with the chemicals, clean, dry where required, and free from cracks or chips.
3. Assemble the flask, distillation head, thermometer, condenser, and receiver using suitable clamps and supports.
4. Check that the apparatus is not sealed into a closed system and that the vapour path and any required pressure-relief route remain unobstructed.
5. Verify that cooling-water hoses are secure, the drain route is safe, and the condenser has adequate water flow.
6. Check the heating equipment, electrical connections, ventilation, and required PPE; prepare suitable spill-response materials.

How to use

1. Place the unit in a suitable laboratory or process area with appropriate ventilation and access control.
2. Add the specified quantity of feed liquid to the distillation flask, leaving adequate headspace for boiling.
3. Add boiling chips or an approved boiling aid before heating when required by the procedure; never add them to a hot, superheated liquid.

4. Secure all joints, clamps, thermometer positions, and the receiving vessel before starting the process.
5. Start condenser cooling water and confirm steady flow and leak-free connections.
6. Switch on the approved heater and increase heat gradually to avoid bumping, foaming, or sudden vapour release.
7. Monitor the temperature, boiling behaviour, condenser performance, and collected distillate throughout operation.
8. Adjust heating only within the procedure's limits and collect fractions in labelled, compatible receiving vessels when required.
9. Do not allow the distillation flask to boil dry; stop heating before the liquid level becomes unsafe.
10. If vapour escapes, cooling water fails, glassware cracks, or abnormal pressure is suspected, stop heating and follow the emergency procedure without touching hot apparatus.
11. At the end of the run, turn off the heater and allow the apparatus to cool while maintaining condenser cooling for as long as the procedure requires.
12. Stop cooling water when vapour generation has ceased and the apparatus has cooled sufficiently, according to the procedure.
13. Transfer and label the collected material as specified, then dispose of residues according to laboratory rules.
14. When fully cool, disassemble the apparatus carefully and clean the equipment using the approved method.

⚠ Safety

- Operate the unit only if trained and authorised for the chemicals and process involved.
- Wear a lab coat, chemical-compatible gloves, and splash goggles; use a fume hood for hazardous or volatile vapours when required.
- Never heat a sealed distillation system; ensure the vapour path is not blocked.
- Keep flammable liquids away from flames and use only an approved heating method for the material.
- Do not touch hot glassware or tighten joints while the apparatus is hot.
- Never leave active distillation unattended unless the approved procedure specifically provides for supervised automatic operation.
- Do not distil unknown mixtures or materials that may decompose dangerously without an approved risk assessment.
- Follow emergency procedures for spills, fire, cooling failure, and chemical exposure.

Care & storage

- Inspect glassware, joints, clamps, tubing, and condenser connections before each use.
- Clean the flask, condenser, and receiving vessels after use using compatible cleaning methods.
- Check cooling-water hoses and electrical heater cables for wear, leaks, or damage.
- Store clean glassware securely and protect it from impact and sudden temperature changes.
- Arrange repair or replacement for cracked glassware, damaged seals, or faulty temperature controls.

Common mistakes to avoid

- Heating the mixture too rapidly or overfilling the distillation flask.
- Operating with blocked vapour paths or a sealed apparatus.
- Starting heating before confirming condenser water flow.
- Allowing the flask to boil dry or handling hot glassware without protection.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.